TN SKILLS

ComicCraft-AI: Comic Story Creator Using Gemini Models

Brainstorming & Ideation Phase

- The brainstorming and ideation phase identifies the problem, target users, possible features, and overall concept of ComicCraft. The team discusses how generative AI can reduce the difficulty of writing comic stories and organize creative ideas into a usable comic format.

- Problem Identification Many beginners have comic ideas but find it difficult to develop a complete story, create meaningful characters, write dialogues, maintain continuity, and divide a story into suitable comic panels. Traditional story-writing also requires time and creative experience.

- Proposed Solution ComicCraft accepts a simple user prompt such as a theme, character idea, genre, setting, or short story concept. Gemini models process the prompt and generate a structured comic story containing a title, characters, plot, scenes, dialogues, narration, and panel descriptions.

- Brainstormed Features Important proposed features include prompt-based story generation, genre selection, character creation, scene generation, dialogue generation, story continuation, editing and regeneration, panel-wise organization, and export or presentation of the generated comic script.

- Target Users The application can be used by students for creative assignments, teachers for classroom activities, aspiring comic writers, content creators, and general users who want to experiment with AI-assisted storytelling.

Requirement Analysis Phase

- Requirement analysis converts the project idea into clear functional and non-functional requirements. It identifies what the user should be able to do and what qualities the system should provide.

- Functional Requirements The system should allow users to enter a story prompt, select a genre or theme, generate a comic story, create character descriptions, generate scenes and dialogues, regenerate selected content, edit generated text, and view the output in an organized format.

- AI Requirements Gemini models should receive structured prompts and return useful creative content. The application should provide appropriate context to maintain character names, settings, tone, and story continuity. Prompt templates can be used to request predictable output sections such as title, characters, panels, narration, and dialogue.

- Non-Functional Requirements The application should be user-friendly, responsive, reliable, maintainable, scalable, and reasonably fast. Generated content should be clearly displayed and errors should be handled gracefully.

- Hardware and Software Requirements A normal computer or smartphone with internet access is sufficient for the client side. Software requirements may include a modern web browser, development environment, application framework, Gemini API access, and a suitable programming language such as Python or JavaScript.

Project Design Phase

- ComicCraft can be designed as a layered application. The user interface collects inputs, the application layer validates requests and constructs prompts, the Gemini service generates content, and the presentation layer formats the response as a comic story.

- The main components are User Interface, Prompt Manager, Gemini API Integration, Story Generator, Character Manager, Scene and Dialogue Generator, Output Formatter, and Error Handler.

- The user opens ComicCraft, enters a story idea, chooses optional settings, submits the request, and waits for the AI response. The application displays the generated title, characters, scenes, narration, and dialogues. The user can edit or regenerate content and finally save or present the result.

- The interface should contain a prompt box, genre selector, character/settings fields, generate button, loading indicator, result area, edit/regenerate controls, and optional export feature.

- User prompt → input validation → prompt construction → Gemini model → generated response → response parsing/formatting → comic display. Errors or incomplete responses are routed to an error-handling component.

Project Planning Phase.

- The main objective is to develop a functional AI-assisted comic story creator that demonstrates practical use of Gemini models in a creative application.
- The work can be divided into requirement collection, UI design, API setup, prompt engineering, backend or application logic, Gemini integration, output formatting, testing, documentation, and demonstration preparation.
- Week 1: brainstorming and requirements. Week 2: system and UI design. Week 3: Gemini integration and core development. Week 4: story formatting and feature refinement. Week 5: testing and bug fixing. Week 6: documentation, presentation, and final demonstration.
- Possible roles include project coordinator, UI developer, application/backend developer, AI prompt and integration developer, tester, and documentation/presentation member.
- In a small student project, one person can perform multiple roles.

Project Development Phase

- The development phase begins by creating the project structure and configuring the required programming environment. Gemini API credentials should be stored securely and should not be hard-coded into publicly shared source code.
- The interface is developed with fields for story prompts and optional parameters. The result section presents the generated comic in a readable structure, with headings for title, characters, panels, narration, and dialogue..
- The application sends the user's request to the Gemini model through the appropriate API or SDK. A structured prompt is created to instruct the model to behave as a comic story creator and return the requested sections.
- The generated response is received by the application and displayed as a comic script. The system can support regeneration when the user is not satisfied with a particular result.
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Project Testing Phase

- Testing verifies that ComicCraft functions correctly, produces useful outputs, handles invalid inputs, and provides a stable user experience.
- Individual functions such as input validation, prompt construction, API response handling, and output formatting can be tested separately.
- Integration testing verifies communication between the user interface, application logic, Gemini service, and output display.
- Test cases should verify story generation, genre selection, character generation, panel generation, editing, regeneration, and error messages.
- A small group of users can try the system and provide feedback about ease of use, clarity of instructions, response presentation, and overall workflow.

Project Documentation Phase

- Documentation records the project idea, requirements, design, implementation, testing, and demonstration so that another person can understand and maintain the system.
- The final documentation should include title page, abstract, introduction, problem statement, objectives, existing and proposed system, requirements, architecture, modules, implementation details, testing, results, limitations, future enhancements, conclusion, and references.
- Screenshots can be included for the home page, prompt input, settings, generated comic output, regeneration/editing, and final result.
- The results section should explain how ComicCraft converts a simple user idea into a structured comic story using Gemini models.
- AI-generated stories may vary between requests and may occasionally contain continuity errors, repetitive content, or unsuitable details. Text generation alone does not automatically produce professionally illustrated comic artwork.

Project Demonstration Phase

- The demonstration shows the working ComicCraft application and explains how Gemini models are used in the system
- Start by introducing the problem and objective. Open the application, enter a sample prompt, select a genre, generate the story, explain the title and characters, show the panel-by-panel output, demonstrate editing or regeneration, and conclude with the project benefits and future scope.
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- During the demo, explain the user input, prompt construction, Gemini model interaction, generated story structure, response formatting, error handling, and testing results.
- The final demonstration should clearly show that a user can provide a simple idea and receive a structured comic story through the ComicCraft interface